



C3PO
interferometer

Inspection Data Sheet

Recipient: UCLA Medical Receiving, Physics and Astronomy
Knudsen Hall Room 1-129
Eric Hudson 210 206-1077
475 Portola Plaza
90095 Los Angeles CA

Customer No.: 14019

Serial No.: 15-12

Model: qu2Pi

Description: Demonstrator for
Houng Ou Mandel effect



C-3PO

Detected characteristics

Fiber to Fiber Coupling Efficiency	>75%
Houng Ou Mandel Dip Visibility	79,00%

Dr. Sebastian Weuerth
04.12.14



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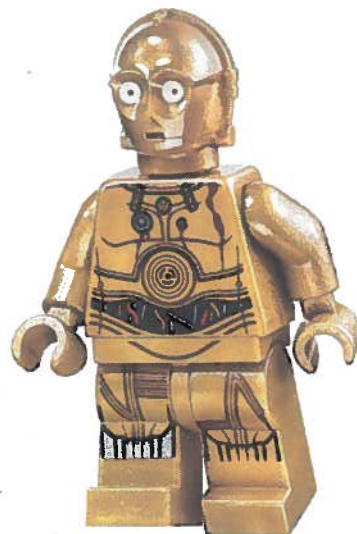
0032

Model:

quED

Description:

Entanglement Demonstrator –
Source of Polarization Entangled Photon Pairs



C - 3P0

Detected characteristics

	Single 0		Single 1	
Dark-count rates	1125	s ⁻¹	1094	s ⁻¹
Single-count rates*	78000	s ⁻¹	71000	s ⁻¹

Coincidence count rate*	5163	s ⁻¹
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Entanglement*	H/V – basis		Diag. - basis	
	> 99,6	%	> 94,6	%

Bell inequality†	2.69	±	0.01
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* laser diode at op. current, irises fully opened, polarizers removed from the setup, twin detection module used

* measured as the visibility of correlation curves in two complementary bases (corrected for random coincidence detections)

† Φ^+ -state analyzed, integration time of 10s per angular setting (not corrected for random coincidence detections)

Sebastian Nauerth

07.12.15

Basic Specifications

Pump Laser Diode		Current [mA]	Power [mW]
	threshold	27.4	1.0
		31.5	6.0
		33.3	8.0
	operating	35.0	10.0
		36.6	12.0
		37.6	14.0
	maximum	50.0	29.0
λ_{peak}		0.0	nm
$T_{\text{operating}}$		25.0	°C
Conversion crystals	material: Beta-barium borate (BBO, β -BaB ₂ O ₄)		
	phase matching: type I at $\Theta = 29.3^\circ$ ($\Phi = 0.0^\circ$)		
	transmission at op. wavelength: $\approx 91\%$		
Polarizers	contrast: $> 1000:1$		
	transmission at op. wavelength: $\approx 80\%$		
Source dimensions	30 x 45 x 10 cm ³		
Measuring Conditions	T_{ambient}	25	°C
	humidity	0	%

Measured data: Bell (CHSH) inequality

Setting	Pol 1	Pol 2	Coinc. [s ⁻¹]
C(α, β)	0°	22.5°	21896
C(α, β s)	0°	112.5°	5628
C(α s, β)	90°	22.5°	4228
C(α s, β s)	90°	112.5°	20918
C(α' , β)	45°	22.5°	20398
C(α' , β s)	45°	112.5°	2947
C(α' s, β)	135°	22.5°	2637
C(α' s, β s)	135°	112.5°	23284
C(α, β')	0°	65.5°	4665
C(α, β' s)	0°	157.5°	25934
C(α s, β')	90°	65.5°	24116
C(α s, β' s)	90°	157.5°	3082
C(α' , β')	45°	65.5°	22355
C(α' , β' s)	45°	157.5°	6514
C(α' s, β')	135°	65.5°	5752
C(α' s, β' s)	135°	157.5°	20956

		Value	Error
E1	$C(\alpha, \beta) + C(\alpha s, \beta s) - C(\alpha, \beta s) - C(\alpha s, \beta) / C(\alpha, \beta) + C(\alpha s, \beta s) + C(\alpha, \beta s) + C(\alpha s, \beta)$	0.62575	0.00001
E2	$C(\alpha', \beta) + C(\alpha' s, \beta s) - C(\alpha', \beta s) - C(\alpha' s, \beta) / C(\alpha', \beta) + C(\alpha' s, \beta s) + C(\alpha', \beta s) + C(\alpha' s, \beta)$	0.77331	0.00001
E3	$C(\alpha', \beta') + C(\alpha' s, \beta' s) - C(\alpha', \beta' s) - C(\alpha' s, \beta') / C(\alpha', \beta') + C(\alpha' s, \beta' s) + C(\alpha', \beta' s) + C(\alpha' s, \beta')$	0.55859	0.00001
E4	$C(\alpha, \beta') + C(\alpha s, \beta' s) - C(\alpha, \beta' s) - C(\alpha s, \beta') / C(\alpha, \beta') + C(\alpha s, \beta' s) - C(\alpha, \beta' s) - C(\alpha s, \beta')$	-0.73192	0.00001
S	E1+E2+E3-E4	2.690	0.006
Std. Deviation	(S-2) / ΔS	108.294	---